Data as a Networked Asset

A Brief Summary

Link to the full paper: https://yeli-macrofinance.com/data_value.pdf

Data is non-rival: one firm's use of data does not diminish its availability to others, allowing the same data to be used by many firms simultaneously (Jones and Tonetti, 2020). Data also exhibits externalities: information collected by one firm on its customers can benefit other firms by revealing consumer preferences and broader economic forces and by serving as training data for prediction models. Non-rivalry, combined with externalities, expands the potential uses of a firm's data, often transcending industry boundaries. We ask: What is the value of a firm's data? How is data collected by one firm shared with others, and what are the scope and network structure of data sharing? How does data sharing affect firms' decision-making and propagate shocks across firms? Which firms are systemically important in the data economy? And what are the impact and unintended consequences of data-sharing regulations?

We trace inter-firm data flows originating from mobile applications (apps), which have become the primary channels for data collection in the economy. Our sample contains 1,031 app-owning public firms that account for more than 60% of the total assets of Compustat firms. Firms' apps collect data and transmit it to software development kits (SDKs) that specialize in data analytics. A crucial function of these "data-sharing SDKs" is merging information from different firms' mobile apps associated with the same consumer into a comprehensive customer profile; by sharing data with the SDKs, firms gain access to consumer insights that combine their own data with that of other firms. This data infrastructure is economically significant: among firms whose revenues come from data-reliant segments, the intensity of data-sharing SDK usage predicts revenues as strongly as R&D intensity, and when SDK providers release major privacy-driven updates that restrict data sharing, the exposed firms experience significantly negative operational and stock-market performance.

Using granular information on app-level SDK installations, we construct the first measure of data-sharing conduits between firms—two firms are connected when a common set of data-sharing SDKs is installed on their apps—and map out the entire network. Data sharing turns out to be a powerful economic linkage. It induces comovement in firms' operational performances, with an economic magnitude more than double that of the product similarity measure of Hoberg and Phillips (2016), and data-connected firms exhibit stock-return correlations that cannot be explained by standard asset-pricing factors or by other common exposures, such as product similarity, supply-chain linkages, and common analyst coverage. The strength of comovement depends on network topology: firms indirectly connected through common data-sharing peers comove over and above the effect of direct connections, and comovement is strongest where the network is locally dense.

We provide direct evidence on a mechanism behind this comovement: targeted advertising. Data-connected firms comove in their digital advertising spending, in how they allocate advertisements across apps and web channels, and in advertising effectiveness—what advertising expenses achieve in customer acquisition. Two features distinguish data sharing from the alternative explanation that common shocks move firms' overall advertising together: the comovement is

absent in non-digital advertising, which relies less on SDK-mediated data, and it is weakened by Apple's privacy policy that disrupts inter-firm data sharing.

Our paper is the first to uncover this new form of economic linkage, and it speaks directly to policy debates on data privacy. With rising consumer awareness of privacy, regulations such as GDPR in Europe and CCPA in California have imposed strict rules on data collection and sharing. In a difference-in-differences framework, we examine the introduction of Apple's App Tracking Transparency (ATT), which allows app users to block identity tracking and thus makes it difficult for SDKs to merge data from different firms. We find that ATT significantly weakens the performance comovement between data-connected firms—an unintended consequence of privacy initiatives. This experiment also validates that firms' overlap in data-sharing SDKs is fundamentally about linkages in data: ATT does not affect the contribution of other inter-firm overlaps, such as product similarity, to performance comovement.

Because comovement ultimately emerges from shocks propagating from one firm to another, we further provide direct evidence using cyberattacks, which destroy firms' data stocks and impair data collection by diminishing the customers' willingness to engage with the mobile apps. When a firm suffers a cyberattack, its data-connected peers experience significantly greater deterioration in operational and stock-market performances than firms that are not data-connected.

These findings raise a measurement question at the heart of the data economy: what is the value of a firm's data? Because data is shared, its value is not confined to the firm where it originates; it includes the present value of the cash flows that the data generates at connected firms, through direct and higher-order spillovers—firm A's data may be shared with firm B, which in turn shares data with firm C—and through their persistent effect over time, as data drives customer engagement and engagement generates further data (Farboodi and Veldkamp, 2025).

To value data as a networked asset, we develop a dynamic network model in which data plays the role of productive capital. Each firm forecasts customer preferences with a precision that depends on both its own data and data from connected peers; more accurate forecasts enhance customer engagement, which boosts customer capital, product demand, and cash flows, and creates new data for the firm and its peers. The model takes the empirically measured network of data-sharing conduits as input, and simulations of the calibrated model replicate the data-induced comovement and the ATT effects. Building on the model, we derive a closed-form valuation formula that embeds the entire network structure of data sharing, the propagation of data through higher-order externalities, and self-perpetuating data growth; absent data sharing, it collapses to the standard Gordon growth model for data valuation. Before ATT, Meta's data is valued at around $780 billion, and the data of all firms in our sample at approximately $1.984 trillion, or 8.4% of U.S. GDP in 2021. The number is larger than the estimate of Bhutani, Ordoñez, and Veldkamp (2026), who value the data that U.S. firms acquire in a year at about 1.5% of GDP, because our number reflects the value of the full data stock (rather than annual data flow) and incorporates spillovers on the data-sharing network. ATT weakened inter-firm data sharing: the ratio of the total value of Meta's data to its own value fell from 3 to 1.15, and that of Alphabet from 6 to 1.33. The framework thus quantifies the value of a firm's data for the whole data economy, its network externality and systemic importance, and how these vary with privacy policies.

Finally, the model features a trade-off distinctive to firms in the data economy: in designing products, a firm balances monetization against customer engagement, and data from connected peers alleviates this intertemporal tension. In equilibrium, product-design decisions exhibit

herding: when one firm prioritizes customer engagement, the data it generates allows its connected peers to do the same without significantly sacrificing current profits. Empirically, a firm's product-design choices are strongly influenced by those of its data-connected peers, even after accounting for product overlap and supply-chain linkages, and this herding weakens after ATT. These results provide an out-of-sample validation of the model—herding is a prediction of the same mechanism that generates performance comovement, and the model's equilibrium dynamics align with the data even though these moments are not targeted in the calibration—and they complete the account of ATT's impact: beyond reducing the value of data as a networked asset, the policy weakened the complementarity in firms' data investment.

References

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